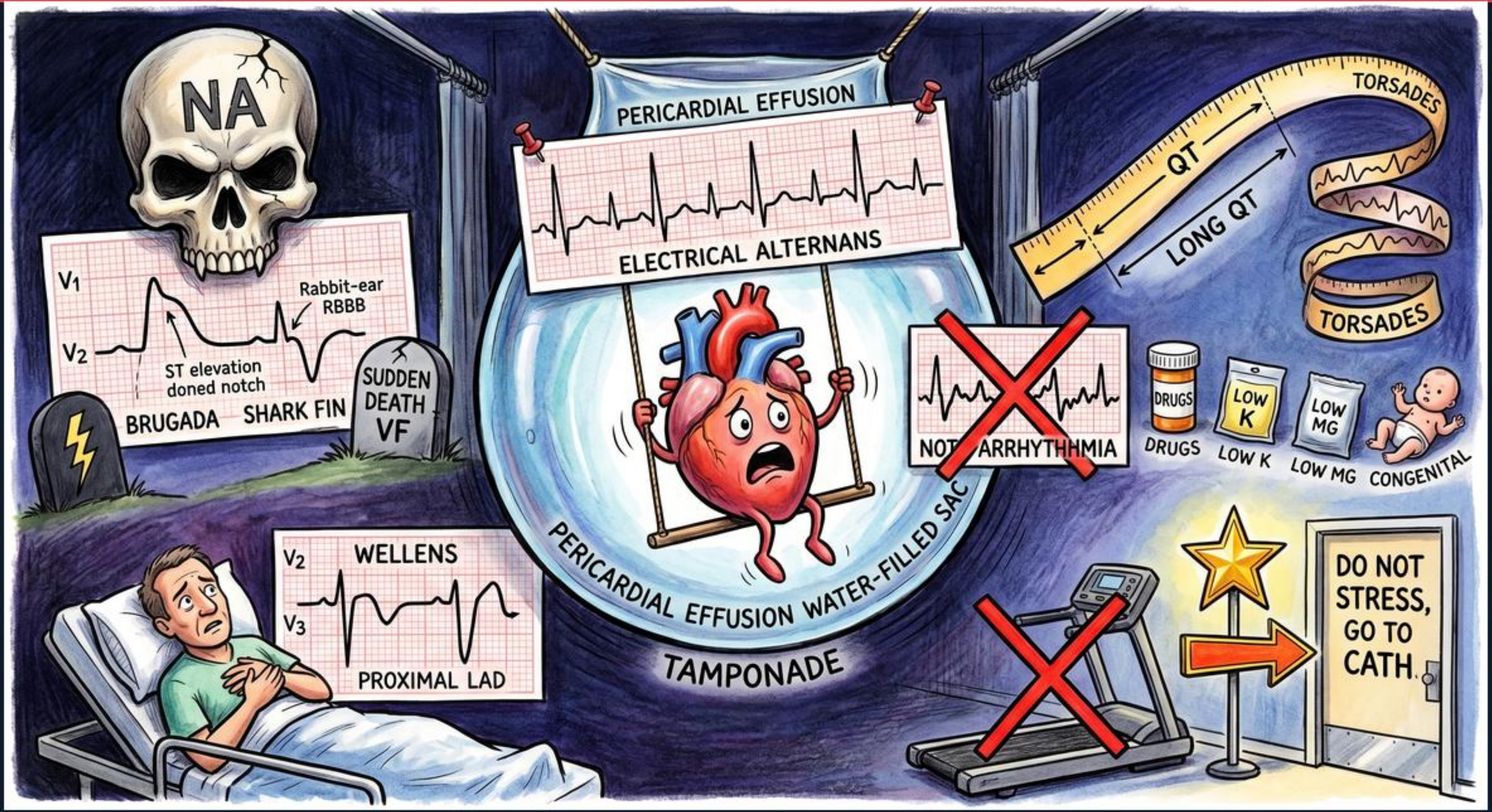
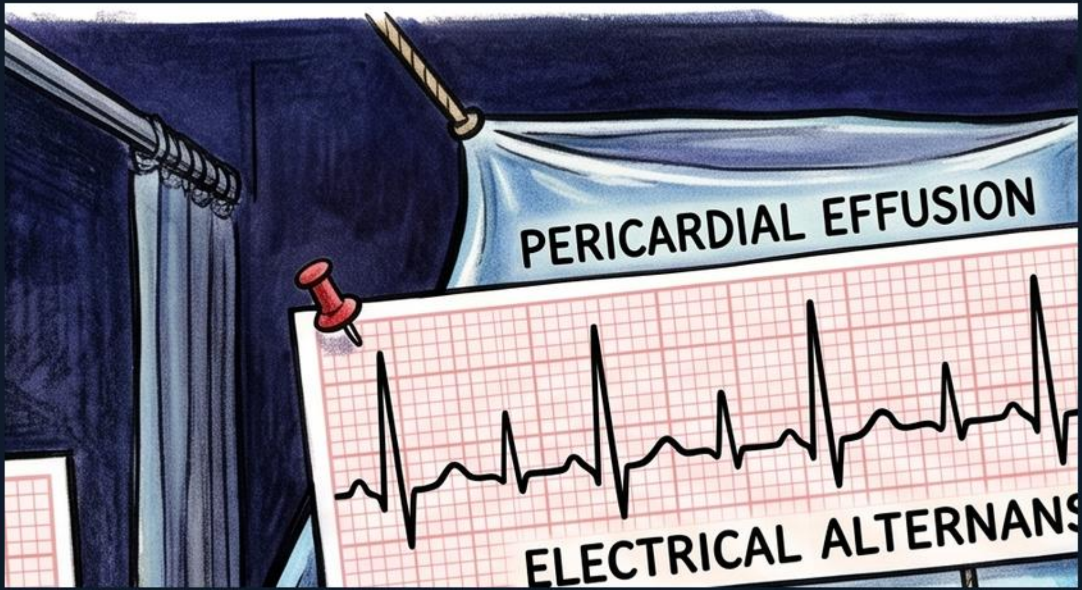
Cardiac Exam — ECG: Danger Patterns



15 symbols · sketch walkthrough



Electrical alternans strip

WHAT YOU SEE

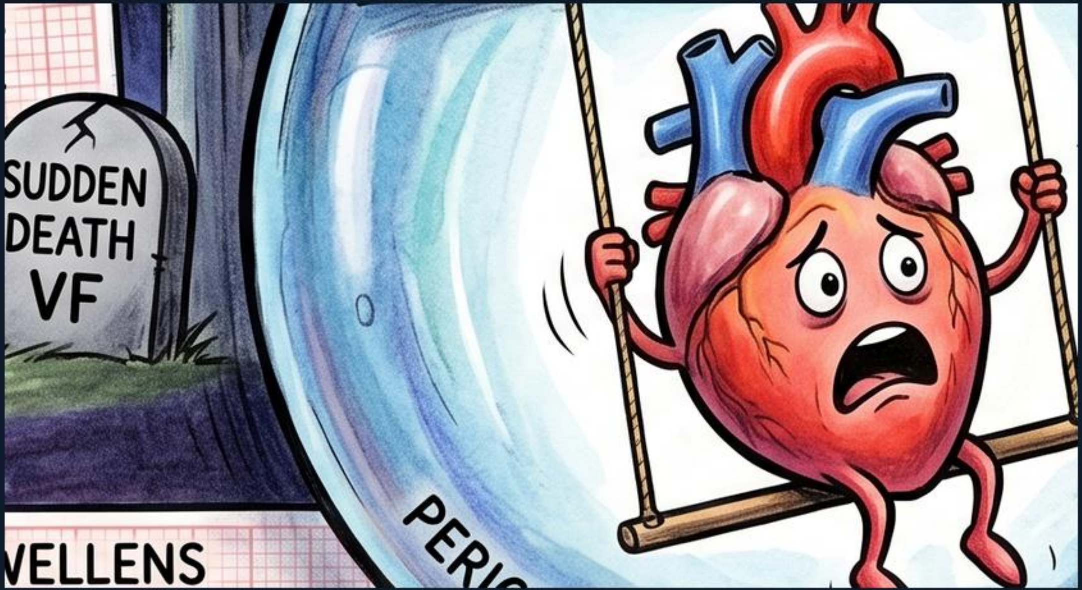
ECG strip labeled 'PERICARDIAL EFFUSION' / 'ELECTRICAL ALTERNANS' where QRS complexes alternate tall, short, tall, short.

WHAT IT ENCODES

Electrical alternans is a beat-to-beat alternation in QRS amplitude caused by the heart physically swinging back and forth within a LARGE pericardial effusion. Each beat the heart sits at a different distance from the electrodes, so the voltage changes. It is a red flag for impending cardiac tamponade, not a primary rhythm problem.

THE BOARD TRAP

Calling electrical alternans an arrhythmia is the classic error — the rhythm is regular; it signals a big effusion/tamponade.



Swinging heart in water-filled sac

WHAT YOU SEE

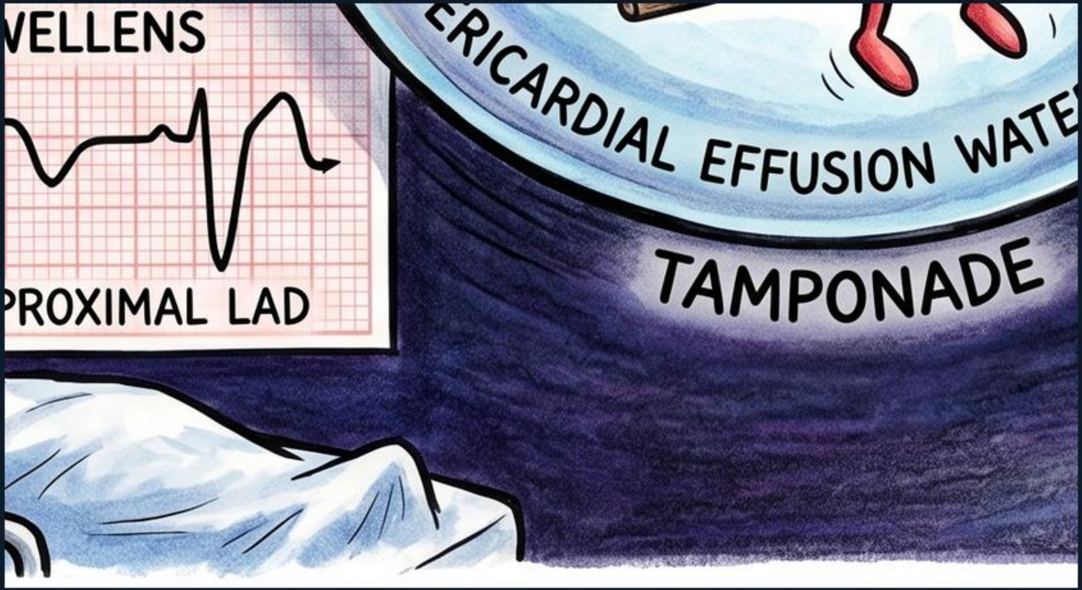
A frightened cartoon heart on a swing inside a translucent water-filled balloon labeled 'PERICARDIAL EFFUSION WATER-FILLED SAC.'

WHAT IT ENCODES

The heart on a swing in a fluid-filled sac is the mechanism of electrical alternans: a large pericardial effusion lets the heart oscillate freely each beat. As fluid accumulates under pressure, diastolic filling is compromised, producing tamponade physiology. Echo, not ECG, confirms the effusion and tamponade.

THE BOARD TRAP

The swinging heart is a mechanical effusion sign — don't attribute the changing voltage to electrode artifact or a conduction defect.



TAMPONADE label

WHAT YOU SEE

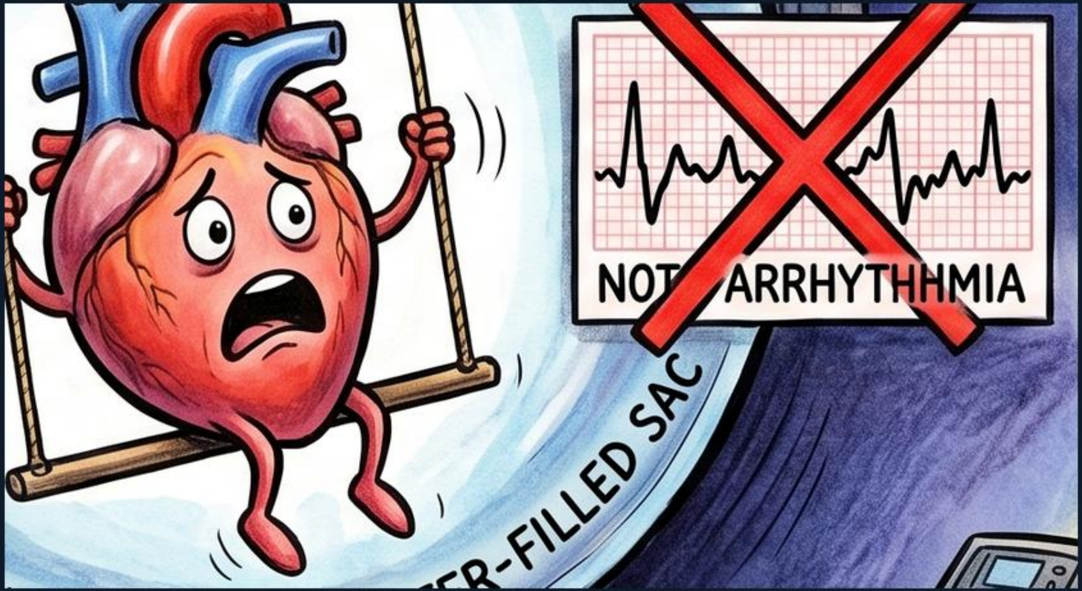
Bold text 'TAMPONADE' at the base of the water-filled sac.

WHAT IT ENCODES

Tamponade is the feared end-point of a large/rapid pericardial effusion: rising intrapericardial pressure collapses the right heart in diastole, dropping cardiac output. Clinically Beck's triad (hypotension, muffled heart sounds, distended neck veins) plus pulsus paradoxus. Electrical alternans on ECG is a supportive clue; treatment is pericardiocentesis.

THE BOARD TRAP

Tamponade is a clinical/echo diagnosis — a normal-sized ECG voltage does not rule it out, and electrical alternans is only sometimes present.



NOT ARRHYTHMIA crossed-out

WHAT YOU SEE

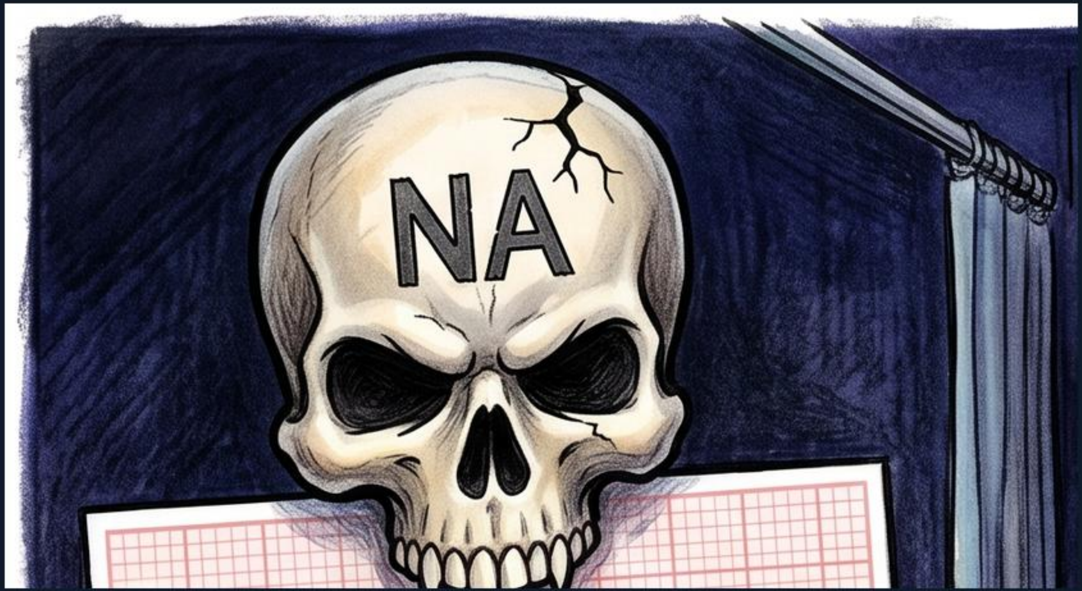
A short ECG segment with a big red X and the words 'NOT ARRHYTHMIA.'

WHAT IT ENCODES

This crossed-out strip reinforces the teaching trap: electrical alternans is NOT a dysrhythmia. The underlying rhythm (usually sinus) is regular; only the QRS height alternates because of the swinging heart. The correct next step is echocardiography to assess effusion and tamponade, not antiarrhythmic therapy.

THE BOARD TRAP

Treating electrical alternans with rate/rhythm control wastes time — look for and drain the pericardial effusion.



NA skull (sodium channel)

WHAT YOU SEE

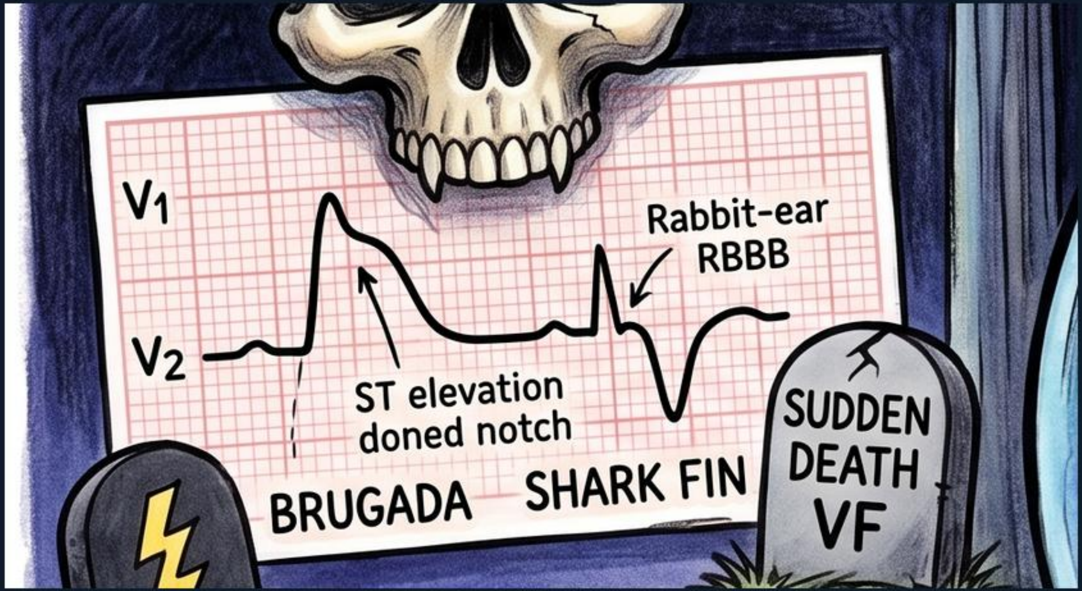
A skull labeled 'NA' in the upper-left corner above the Brugada tracing.

WHAT IT ENCODES

The 'NA' skull marks Brugada syndrome as a SODIUM-channelopathy (loss-of-function SCN5A mutations) carrying a risk of sudden cardiac death. Reduced sodium current alters right-ventricular epicardial repolarization, generating the coved ST pattern. Fever, certain drugs and vagal tone can unmask it.

THE BOARD TRAP

Brugada is a channelopathy, not a structural disease — a structurally normal heart with this ECG still carries sudden-death risk.



Brugada strip — coved ST V1-V2

WHAT YOU SEE

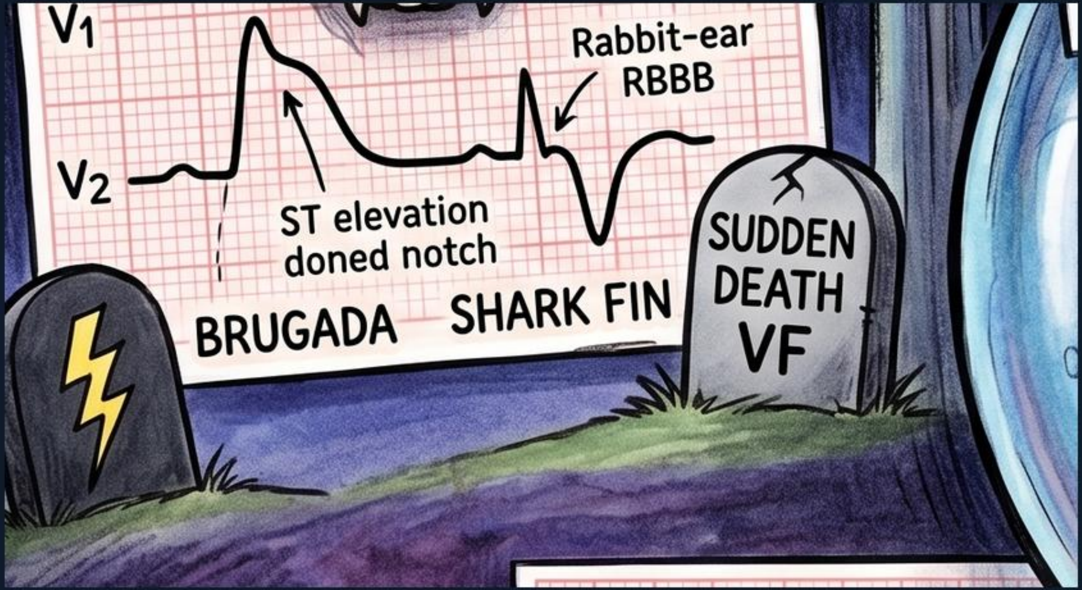
ECG strip labeled 'BRUGADA SHARK FIN' showing leads V1-V2 with ST elevation, a domed notch, and a partial RBBB-like rabbit ear.

WHAT IT ENCODES

Brugada (type 1) shows a coved, downsloping 'shark-fin' ST-segment elevation in V1-V2 followed by a negative T wave, often with an incomplete RBBB pattern. It reflects the sodium channelopathy and identifies patients at risk of polymorphic VT/VF and sudden death. An ICD is the definitive therapy for high-risk patients.

THE BOARD TRAP

Don't dismiss coved V1-V2 elevation as ordinary RBBB or early repolarization — the coved 'shark fin' morphology is Brugada.



SUDDEN DEATH / VF tombstone

WHAT YOU SEE

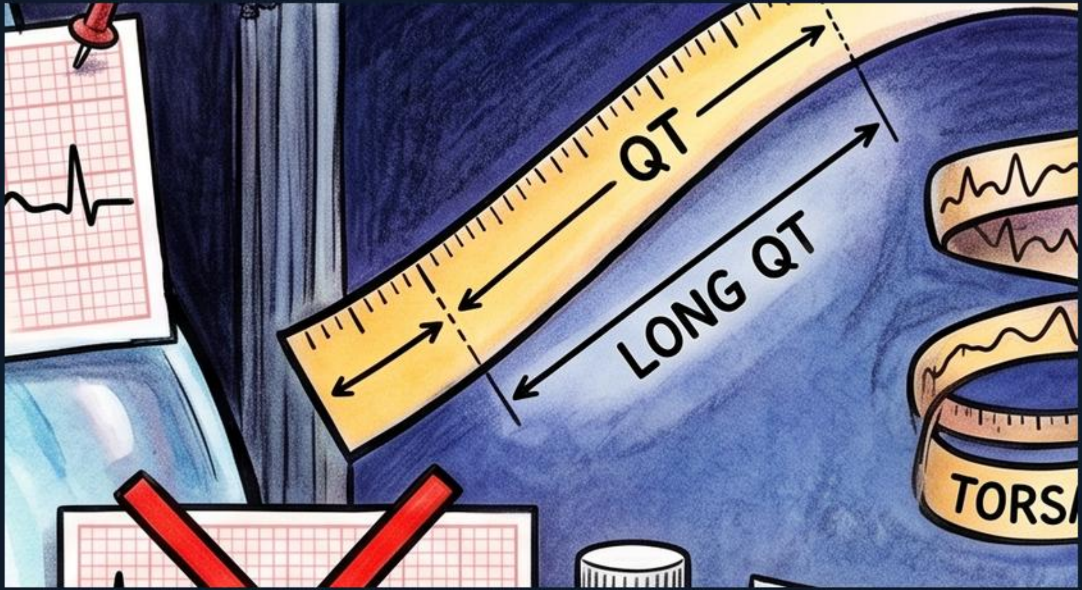
A tombstone-style graphic reading 'SUDDEN DEATH VF' beside the Brugada strip.

WHAT IT ENCODES

Brugada's danger is sudden cardiac death from ventricular fibrillation, often during sleep or fever. Risk stratification considers spontaneous type-1 pattern, prior syncope or arrest, and inducibility. High-risk patients receive an implantable defibrillator; quinidine is an adjunct.

THE BOARD TRAP

An asymptomatic incidental Brugada pattern still needs evaluation — don't assume a well-appearing patient is automatically low risk.



Long QT ruler

WHAT YOU SEE

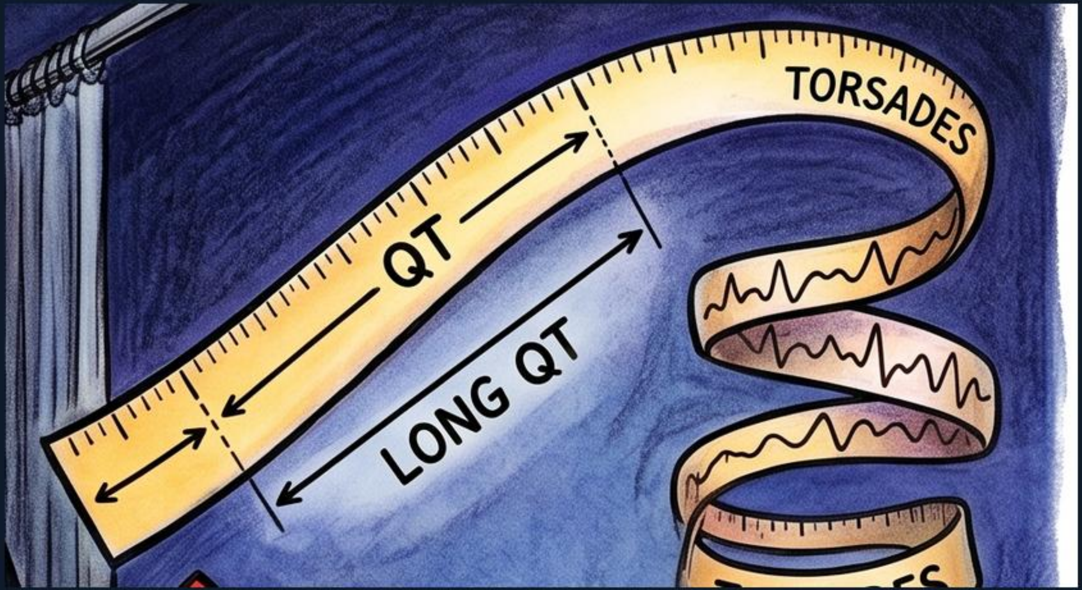
A yellow measuring-tape/ruler stretched long across the top right, labeled 'QT' and 'LONG QT.'

WHAT IT ENCODES

The over-long ruler depicts a prolonged QT interval — delayed ventricular repolarization. A long QT (corrected QTc) predisposes to early afterdepolarizations that trigger torsades de pointes. Measure QT in lead II or V5 and correct for rate (QTc).

THE BOARD TRAP

Always use the corrected QTc, not the raw QT — a long raw QT at a slow heart rate may be normal once rate-corrected.



Torsades de pointes twists

WHAT YOU SEE

Two helical, ribbon-twisting ECG tracings labeled 'TORSADES' that spiral around a central point.

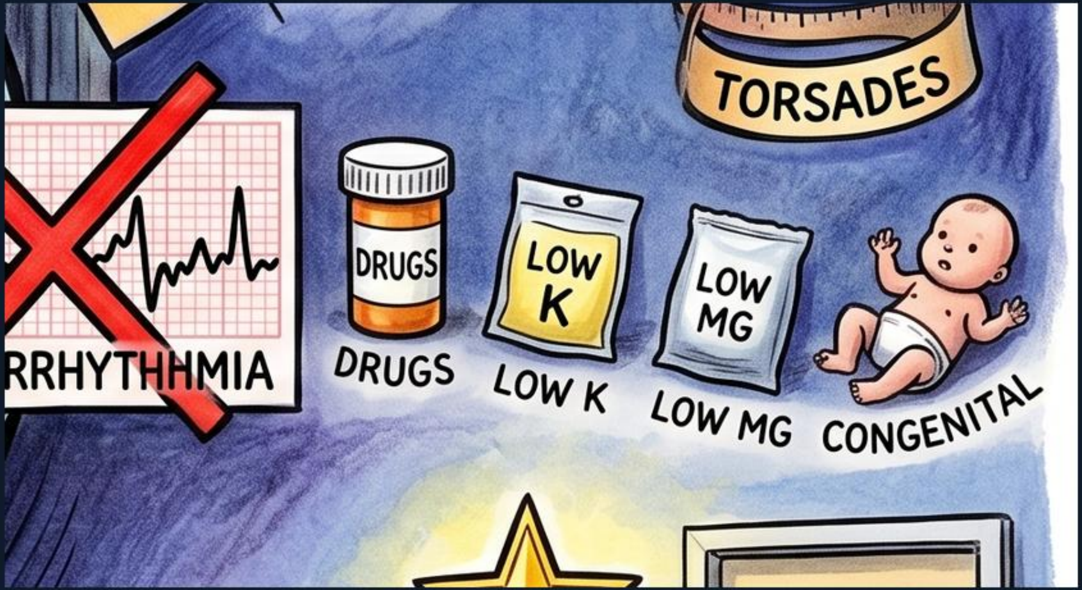
WHAT IT ENCODES

Torsades de pointes is a polymorphic ventricular tachycardia in which the QRS complexes twist around the baseline ('twisting of the points'). It arises from a prolonged QT and can degenerate into VF. Acute treatment is IV magnesium, correcting electrolytes, and overdrive pacing or isoproterenol for bradycardia-dependent forms.

THE BOARD TRAP

Torsades requires a long QT context — treat with magnesium and remove the offending cause; standard antiarrhythmics that prolong QT can worsen it.

Long-QT causes — drugs / low K / low Mg / congenital



WHAT YOU SEE

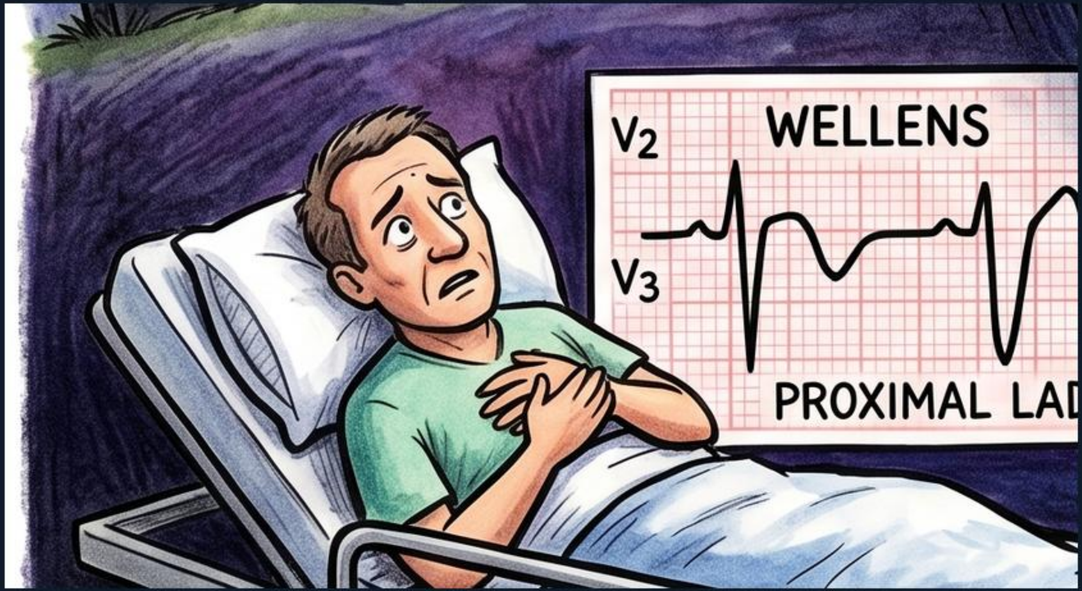
A row of cards reading 'DRUGS,' 'LOW K,' 'LOW MG,' and a baby beside a door labeled 'CONGENITAL.'

WHAT IT ENCODES

The reversible and inherited causes of long QT: QT-prolonging DRUGS (antiarrhythmics, macrolides, antipsychotics, methadone), HYPOKALEMIA, HYPOMAGNESEMIA, and CONGENITAL long-QT syndromes (e.g., Romano-Ward, Jervell-Lange-Nielsen). Always scan the med list and replete potassium and magnesium.

THE BOARD TRAP

Don't fixate on a single cause — long QT is frequently multifactorial (a drug plus low K plus low Mg stacking together).



Wellens patient — chest pain

WHAT YOU SEE

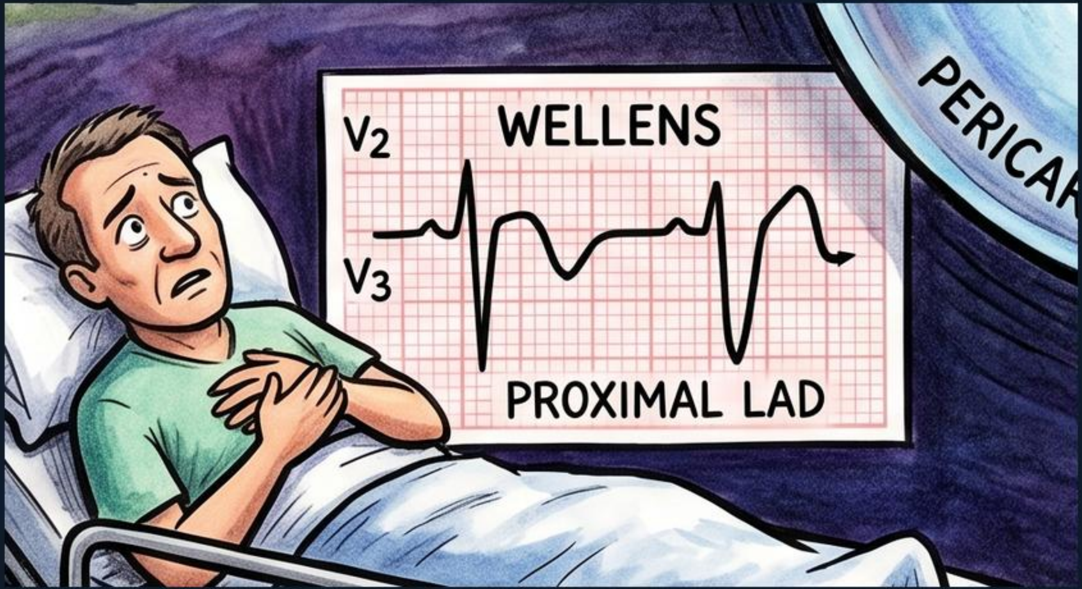
A man lying in a hospital bed clutching his chest, beside the Wellens ECG card.

WHAT IT ENCODES

The bedbound chest-pain patient sets up Wellens syndrome: a patient who had recent anginal chest pain, now often pain-free, whose ECG reveals critical proximal LAD disease. The T-wave changes are typically captured during the pain-free interval. This is a pre-infarction state demanding prompt angiography.

THE BOARD TRAP

A pain-free patient with a benign exam can still be a Wellens — the resolved pain is reassuring but the ECG is the warning.



Wellens strip — biphasic/deep T in V2-V3

WHAT YOU SEE

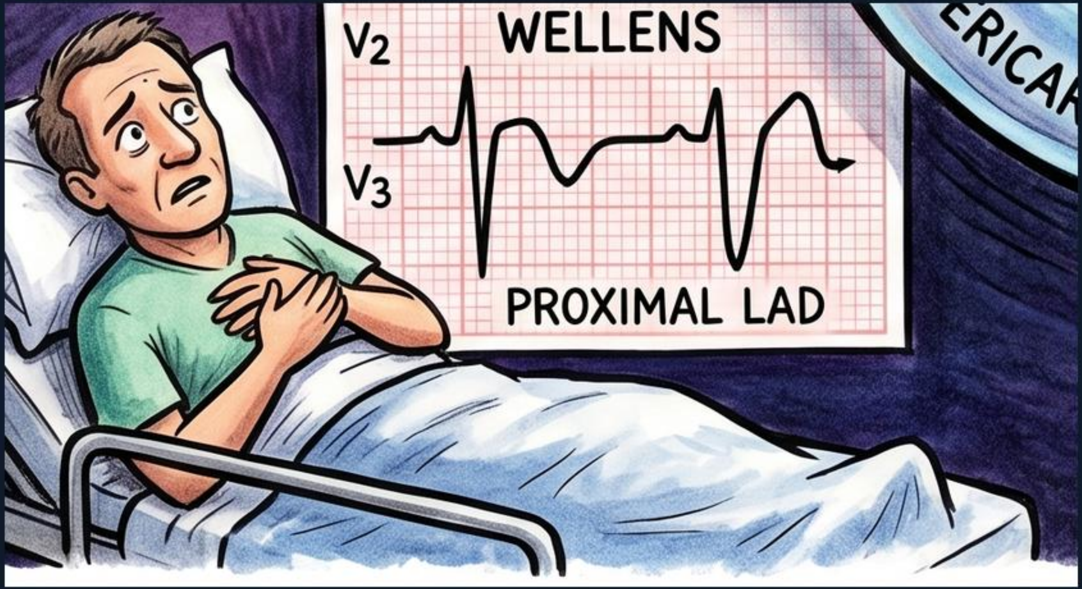
ECG card labeled 'WELLENS' with leads V2-V3 showing deep symmetric (and biphasic) T-wave inversions.

WHAT IT ENCODES

Wellens pattern is deep symmetric or biphasic T-wave inversions in V2-V3 (Type A biphasic, Type B deeply inverted) with little or no ST elevation and preserved R waves. It indicates critical stenosis of the proximal LAD and high risk of imminent anterior MI. The correct path is cardiac catheterization.

THE BOARD TRAP

Subtle V2-V3 T changes in a pain-free patient are easily dismissed — recognizing them as Wellens prevents a missed proximal LAD lesion.



PROXIMAL LAD label

WHAT YOU SEE

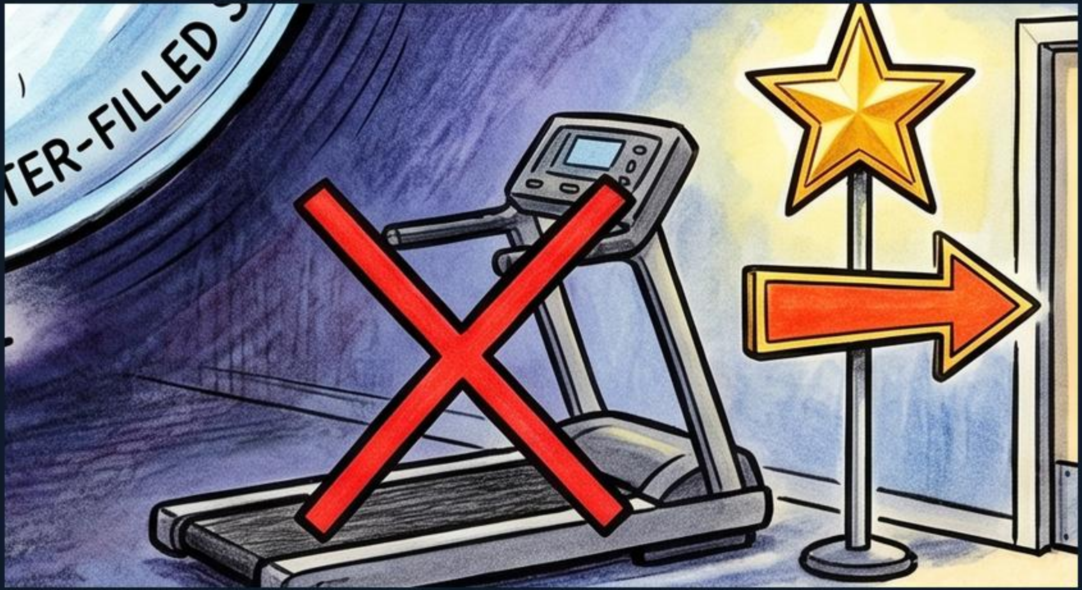
Text 'PROXIMAL LAD' beneath the Wellens ECG card.

WHAT IT ENCODES

Wellens localizes the culprit to the PROXIMAL left anterior descending artery. A tight proximal LAD lesion threatens a large anterior territory, so untreated Wellens frequently progresses to extensive anterior MI within days. Definitive management is angiography and revascularization.

THE BOARD TRAP

Because the lesion is proximal LAD, the territory at risk is large — under-triaging Wellens risks a massive anterior infarct.



Treadmill crossed out (no stress test)

WHAT YOU SEE

A treadmill with a large red X over it on the lower right.

WHAT IT ENCODES

The crossed-out treadmill is the central Wellens rule: do NOT stress-test a Wellens pattern. Provoking ischemia against a critical proximal LAD stenosis can trigger acute infarction or arrest. These patients bypass stress testing and go to invasive angiography.

THE BOARD TRAP

Ordering a stress test for a Wellens ECG is the dangerous trap — it can precipitate the very MI you are trying to prevent.

DO NOT STRESS, GO TO CATH sign

WHAT YOU SEE

A gold star and a sign reading 'DO NOT STRESS, GO TO CATH' on the far right.

WHAT IT ENCODES

This sign states the disposition for Wellens: skip stress testing and proceed directly to cardiac catheterization for the critical proximal LAD stenosis. Early angiography with PCI prevents the impending anterior MI. The gold star marks it as the must-know exam answer.

THE BOARD TRAP

The board-favored wrong answer is 'exercise stress test' — for Wellens the only safe choice is straight to the cath lab.

Summary — every symbol

1. Electrical alternans strip

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2. Swinging heart in water-filled sac

The heart on a swing in a fluid-filled sac is the mechanism of electrical alternans: a large pericardial effusion lets the heart oscillate freely each beat

3. TAMPONADE label

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8. Long QT ruler

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9. Torsades de pointes twists

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